

Robbie Hao

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SUMMARY

M.S. Statistics candidate with applied machine learning experience building scalable predictive pipelines and evaluation workflows on large datasets. Developed reinforcement learning agents for optimal execution and evaluated their robustness across changing liquidity and volatility conditions. Expert in large-scale data processing, advanced statistical modeling, and applying rigorous research methodologies to solve complex real-world problems through machine learning innovation.

EDUCATION

Master of Statistical Science — Duke University	2025–Expected May 2027
B.A. in Economics and Statistics — Dongbei University of Finance and Economics	2021 – 2025

PROFESSIONAL EXPERIENCE

Fubon Fund Quantitative Researcher Intern	Jan.–Mar. 2024; Jul.–Sept. 2024
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- Built a vectorized Python factor-research pipeline implementing 101 Alpha-style signals across ~5,000 equities and 12 years of data, cutting a full-universe backtest from 6 hours to 20 minutes and enabling same-day iteration on new signal ideas.
- Designed a multi-stage screening framework (rank IC/ICIR, pairwise correlation filtering at $|\rho| < 0.6$, Lasso and RF selection) that narrowed 101 raw signals to 14 low-redundancy candidates with mean rank IC of 0.035 and ICIR above 0.45.
- Built long-short portfolios showing 1.6 gross Sharpe decaying to 0.4 net of 20bps cost at 45% turnover, flagging short-horizon signals as capacity-constrained.
- Validated with Fama-MacBeth and Fama-French attribution, isolating 5 factors with residual alpha at $t > 2.8$ after size, value, momentum controls.
- Wrote 6 memos on signal stability, turnover sensitivity, and out-of-sample decay; 2 factors advanced to PM evaluation.

RESEARCH

Graduate Research—Duke Reinforcement Learning for Optimal Execution	April 2026 – Present
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- Built a DQN execution agent in ABIDES multi-agent LOB simulation with a 5-dim rolling cost state (spread, book imbalance, price impact, realized slippage, top-5 depth), benchmarked against TWAP and Almgren-Chriss across 8 regimes and 4,800 paired held-out episodes; cut execution cost 34–37% below TWAP in normal conditions and 72% below TWAP / 81% below Almgren-Chriss on a held-out severe-illiquidity regime (paired $t = -4.5$).
- Isolated an axis-asymmetric failure mode in which policies that dominate the liquidity axis underperform TWAP by 30–50 bps on held-out high-volatility regimes; corroborated by a separate 30-policy transfer study (3 regimes $\times$ 5 seeds $\times$ 2 reward designs) where volatility-trained policies paid $\sim 2\times$ cost in calm markets while reverse transfer was near-free.
- Diagnosed reward gaming in which agents under-executed 4–7% of parent orders to exploit a linear completion penalty, distorting cross-regime comparisons by up to 30%; live-book terminal settlement with exact reward folding ($\gamma=1$) cut mean residual inventory from 22% to under 4% and reduced cross-seed dispersion 34–55% under held-out volatility stress.

Research Assistant—The Second Hospital of Dalian Medical University	May 2024–May 2025
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- Built an ML feature-selection and prediction pipeline for high-dimensional lipidomics data (498 features, 140 samples), identifying 10 biomarkers with ~ 0.95 diagnostic AUC.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, SciPy, Matplotlib, PyTorch), R, SQL.

ML/Stats: Random Forest, LightGBM, XGBoost, Lasso/Ridge Regression, Scikit-Learn, TensorFlow, Deep Learning methods, Reinforcement Learning, Bayesian methods

Finance: multi-factor equity models, backtesting, alpha research

Data & Tools: Git, Linux, SLURM (HPC cluster computing), LaTeX, Excel